

EX:NO:26

PROGRAM:

```
def orientation(p, q, r):  
    return (q[1]-p[1])*(r[0]-q[0]) - (q[0]-p[0])*(r[1]-q[1])  
  
def convex_hull(points):  
    if len(points) < 3:  
        return points  
  
    hull = []  
    l = min(range(len(points)), key=lambda i: points[i][0])  
    p = l  
  
    while True:  
        hull.append(points[p])  
        q = (p + 1) % len(points)  
  
        for r in range(len(points)):  
            if orientation(points[p], points[q], points[r]) < 0:  
                q = r  
  
        p = q  
        if p == l:  
            break  
  
    return hull
```

```
# Test
```

```
points = [(1,1),(4,6),(8,1),(0,0),(3,3)]
```

```
print(convex_hull(points))
```

Output

```
[(0, 0), (4, 6), (8, 1)]
```

```
=== Code Execution Successful ===
```